Flight delay project | Viva notes

Project overview and structure - based on the four executed notebooks in notebooks/simple/

1. What is this project about?

Say this: "Our project predicts whether an operated flight will depart at least 15 minutes late, using a simulated cutoff two hours before scheduled departure. We study ATL, DFW and ORD, compare four classifiers, and test whether weather, earlier delay history and scheduled traffic improve on schedules alone. We also examine prediction errors and whether risk scores can prioritize a limited review list."

Target: delayed_departure = 1 when departure delay >=15 minutes; otherwise 0. Cancelled flights and flights without a known delay are excluded from prediction labels. Later-diverted flights can remain when their departure outcome is known. This is binary classification of departure delay; it does not estimate the number of minutes late.

Data and scope: BTS flight records, airport/time-zone reference information and archived origin weather reports. ATL = Atlanta, DFW = Dallas/Fort Worth and ORD = Chicago O'Hare. Other airports can appear as destinations. Earlier source months provide history context.

Comparison	Inputs
A: schedule	26 inputs: airline/airports, distance, scheduled duration and calendar/time representations.
B: richer inputs	A + weather, earlier delay history and scheduled traffic, with selected pressure/nonlinear additions. LR/KNN: 78 inputs; RF: 72; SVM: 66.

These counts are input columns **before** one-hot expansion. The four models are Logistic Regression (LR), K-Nearest Neighbors (KNN), RBF Support Vector Machine (SVM) and Random Forest (RF). The four aims are classifier selection, A/B feature value, error reliability and risk prioritization.

2. How is the project structured?

Location	Role in the project
notebooks/simple/	01 cleaning/EDA -> 02 features -> 03 training -> 04 evaluation.
data/raw/; data/processed/	Archived source records; cleaned/context data used by the workflow.
scripts/; config/	Readable shared decoding, joins and preprocessing helpers; scope/settings. Classifier definitions and fitting calls stay visible in notebook 03.
reports/simple_reproduction/	features.parquet -> models.joblib -> metrics, scores, errors, priority tables and verification.
models/; reports/final_models/	Published fitted models, frozen development/selection evidence and findings.
dashboard/; tableau/	Interactive HTML dashboard and frozen evaluation exports for visual reporting.

Current experiment: final fitting uses 40,000 flights across all of 2025; evaluation reproduces 10,000 July 2026 flights. The four short notebooks reproduce a completed analysis. July is already exposed and is not a new untouched test. The study assumes timely access to weather, earlier outcomes and historical schedules; live availability is unverified.

Sources: README.md; OBJECTIVES.md; notebooks/simple/README.md.

Notebook 01 | Cleaning and EDA

01_cleaning_and_eda.ipynb - C1, C2, ... refer to code cells, counting imports and setup.

C1-C2 | Setup. Find the project root, import the shared helper and load pandas/plotting libraries. No analytical finding; the font-cache notice is harmless.

C3 | Load and inspect. Reads 17 monthly files from November 2024 to March 2026 and keeps the three origins. Output: **1,339,044 scoped rows**, a monthly source table and five example rows. Stable file/line flight IDs and duplicate checks make rows traceable. The scope is three hubs, not the national population.

C4 | Types, conversions, missingness. The frame initially has 15 columns; all seven numeric conversion audits show **zero unparseable values**. Departure delay and the source 15-minute flag each have **27,926 missing values**; displayed identity/schedule fields are complete. Insight: missing outcomes cannot safely be labeled on time. Clean parsing does not mean every label exists.

C5 | Eligibility and target. Creates labels only for noncancelled, known-delay flights; >=15 minutes becomes 1. Ineligible rows stay unlabeled as schedule context. Departure delay constructs y and never enters the current flight's predictors.

C6 | Timestamps and original splits. Converts HHMM to minutes and origin-local times to UTC, then subtracts two hours. Output below distinguishes all schedules from eligible queries. Insight: context and exclusions must stay separate from model queries.

Original role	Scheduled	Eligible
Context: Nov-Dec 2024	153,143	0
Train: Jan-Sep 2025	712,534	698,670
Validation: Oct-Dec 2025	244,781	239,747
Original test: Jan-Mar 2026	228,586	219,748

Viva distinction: These are the original period flags and full-population counts. The final 40k all-2025 fit and 10k July evaluation are selected separately in notebook 02.

C7 | Describe, skewness and outlier checks. EDA uses the **698,670 eligible Jan-Sep 2025 flights**. Median departure delay is **-1 minute**, mean 17.13, range -60 to 2,364; skewness 8.11 shows a strong right tail. Distance/duration skewness is 1.51/1.46. IQR flags: 20,787/27,641; three-SD flags: 2,762/4,093; above-p99: 6,431/6,827. IQR uses quartile fences; three-SD uses mean/spread; p99 marks the upper 1%. Different rules flag different tails; a flagged value is not automatically invalid.

C8 | Treatment and reconciliation. Only nonpositive distance/duration becomes missing, with invalid-value flags. Genuine early departures and extreme delays stay. Output confirms the cleaned handoff matches on **1,339,044 rows and 24 columns**, including missingness.

C9 | Class balance. 518,143 on-time vs 180,527 delayed, or about 74.16%/25.84%. The unequal classes explain why accuracy alone can hide missed delays. The 10k plot sample is for display; the counts use all eligible training rows.

C10-C11 | Histogram and distance boxplot. The histogram clusters near zero with a long positive tail; negative delays mean early departure. Distance distributions overlap heavily between classes, so distance alone does not clearly separate delayed flights.

C12 | Airport, airline and correlation tables. Delay rates: **DFW 29.13%, ATL 24.26%, ORD 24.17%**. Airline examples range from YX 16.00% to F9 34.58%, with unequal counts. Distance-duration correlation is **0.986**, indicating strong redundancy. These are descriptive associations, not causal airline rankings.

C13 | Departure-hour chart. Rates are around 10% at 05:00-06:00 and generally increase toward roughly 34%-39% in the late evening. Schedule time is a useful candidate feature; the chart alone does not establish why delays rise or how accurate a model will be.

Source: notebooks/simple/01_cleaning_and_eda.ipynb, saved code-cell outputs C1-C13.

Notebook 02 | Feature engineering

02_feature_engineering.ipynb - turn clean context into cutoff-safe, unscaled model inputs.

C1-C2 | Setup. Imports the helper, NumPy, pandas and plotting libraries; no data insight.

C3 | Context and shared queries. Extends context through April-July 2026, then selects **40,000 frozen all-2025 fitting flights** and **10,000 July 2026 flights**. Seed-42 proportional month-by-outcome sampling preserves natural class prevalence. All A/B models use identical flight IDs, so differences do not come from different samples.

C4 | Calendar and cyclic time. Adds weekday, time bands, daily/monthly sine-cosine coordinates, second harmonics and airport-time interactions. Example: 638 minutes (10:38) gives daily sin 0.3502 and cos -0.9367. The daily angle is 2*pi*minutes/1440; month uses 2*pi*(month-1)/12. Midnight stays continuous; harmonics allow multiple daily peaks, and interactions allow airport-specific time patterns.

C5 | Match weather. Uses the latest origin observation at least **20 minutes before cutoff**, and at most **180 minutes old at cutoff**. Example matched ages are 46, 53, 73, 73 and 77 minutes. If cutoff is 12:00, 11:40 can qualify but 11:45 cannot. Identical reports are deduplicated, approved conflicting versions excluded, and unmatched flights retained. This prevents later observations from leaking into earlier predictions.

C6: fitting weather audit	Missing	What the output means
Temperature	0.00%	Mean 63.48 F; range -8 to 102 F.
Wind speed	0.0225%	Mean 8.26 knots; median 8; maximum 32.
Gust speed	80.83%	Observed gusts average 22.54 knots; unknown is not zero.
Ceiling	48.53%	Many reports lack a usable ceiling value.
Rain indicator	90.50%	Unknown condition codes cannot be read as no rain.

C6 insight: A matched report can still have missing measurements. Keep missingness explicit and learn imputation from fitting rows. Trace precipitation uses zero plus a trace flag; dewpoint spread, wind vectors, freezing/wet indicators and prior-three-hour summaries add weather context. Overlapping precipitation totals are not summed.

C7 | Wind boxplot. A 5,000-flight fitting sample shows both outcome classes with median wind near 8 knots and substantial overlap. Wind alone is not a reliable separator; this plot does not measure its added value in a trained model.

C8 | Earlier history and traffic. History uses a **28-day event window ending seven days before cutoff**: [cutoff-35d, cutoff-7d). Compute origin, airline/origin, route and flight-number delay rates plus counts/absence flags. Subgroup smoothing is **(delays + 20*origin_rate)/(n+20)**; 2/5 with origin rate .30 becomes .32. This stabilizes sparse groups. Traffic counts all origin schedules in the hour before scheduled departure, including later cancellations; it measures scheduled demand, not actual traffic.

C9 | History-risk chart. Eight equally sized risk bins show observed fitting delay rates rising broadly from **18.28% to 38.16%**, with a middle-bin dip. Earlier history contains useful association, but the relation is neither perfectly monotonic nor causal.

C10 | Pressure and wind context. Parses METAR altimeter settings before RMK (A2992 = 29.92 inHg), three-hour pressure change/range, variable wind and gust excess. Examples include 30.08 inHg with +0.04 change and 30.03 with -0.01. These represent weather dynamics; the example rows alone do not establish predictive improvement.

C11 | Six nonlinear formulas. Temperature: ((T-32)/10)^2; wind: w^2; dewpoint spread: spread^2; precipitation: log(1+p); ceiling: log(1+c); visibility: 1/(1+v). Squares expose curvature, logs compress long scales and inverse visibility emphasizes poor visibility. Missing inputs stay missing; these are frozen features, not new selections.

C12 | Save features. Saves **50,000 unscaled rows** to features.parquet: 40k refit + 10k July. Labels/audit fields are present for checking, so notebook 03 selects X with explicit allowlists. Independent verification matches all 78 selected features and their missing values on all 50k rows.

Sources: notebook 02, C1-C12; features.parquet (history-bin rates); verification.json.

Notebook 03 | Model training

03_model_training.ipynb - four classifiers x two feature sets = eight fitted pipelines.

C1-C2 | Setup. Loads the helper, timer and four scikit-learn classifiers.

C3 | Training cohort and y. Loads the shared 40k all-2025 refit sample and separates the binary target. The displayed development cohorts preserve time order:

2025 training period	Threshold calibration	Later scoring period
Jan-Mar: 40k sampled flights	April	May-June
Jan-Jun: 40k sampled flights	July	August-September
Jan-Sep: 40k sampled flights	October	November-December

The final all-2025 refit overlaps development; it is not independent validation. "Calibration" here means choosing a decision threshold, not calibrating probabilities.

C4 | Development evidence and thresholds. Mean development F1: LR **0.4835**, RF 0.4816, KNN 0.4742, SVM 0.4472. LR/B was selected by mean F1 with AP as tie-break. Eight separate A/B thresholds came from October maximum-F1 scoring; the largest threshold breaks ties. LR/B uses **0.183914**, rather than a default 0.5. These choices transfer unchanged to the all-2025 refits.

What preprocessing happens before each fit?

The helper returns X in the frozen feature order and an **unfitted Pipeline**. ColumnTransformer applies the frozen median/imputation, missingness-indicator and one-hot-encoding recipes. Continuous inputs are scaled for LR/KNN/SVM; RF omits scaling. Medians, scales and categories are learned only when fitting on the 40k training rows. This prevents test information entering preprocessing and avoids treating airline codes as numeric distances.

Code cell / model	Frozen settings and explanation
C5: Logistic Regression	C=0.01, max_iter=3000. A regularized linear decision rule over encoded inputs. Smaller C means stronger regularization. B uses 78 inputs.
C6: KNN	101 Euclidean neighbors, weighted by distance. Similar training flights vote; scaling matters. B uses 78 inputs. Prediction can be slower than fitting.
C7: RBF SVM	C=3, gamma=scale. RBF similarity supports a nonlinear boundary. B uses 66 inputs; its decision scores are margins, not probabilities.
C8: Random Forest	200 trees, depth 16, minimum leaf 5, sqrt feature sampling and balanced class weights. Combines tree decisions; B uses 72 inputs.

C9 | Save fits and runtime. Saves eight fitted models plus frozen thresholds to models.joblib. Recorded B fitting times: LR **0.38 s**, KNN 0.19 s, SVM 248.71 s and RF 7.71 s. These are run-specific timings. All four visible .fit calls execute for A and B; the helper never fits a classifier.

Results to remember for the next notebook

July 2026	A F1	B F1	B AP	B precision	B recall
LR (selected)	0.5012	0.5395	0.5665	0.3810	0.9241
KNN	0.5132	0.5454	0.5324	0.4007	0.8535
RBF SVM	0.4866	0.5342	0.5130	0.4349	0.6924
Random Forest	0.5405	0.5429	0.5404	0.3926	0.8796

Why keep LR? Choosing again because KNN has higher July F1 would tune on evaluation outcomes. LR remains the validation-selected model.

Sources: notebook 03, C1-C9; notebook 04 metrics. All paths are relative to the project.

Notebook 04 | Evaluation and findings

04_evaluation_and_findings.ipynb - explain each output through the four project objectives.

C1-C3 | Setup and evaluation data. Loads saved fits and the same sorted 10k July flights. **3,215 are delayed (32.15%)**. Predictions use frozen thresholds; no July tuning.

C4-C5 | O1: metrics table and A/B F1 chart. The eight model rows are summarized on page 4. Always-on-time has **67.85% accuracy but delayed F1=0**; always-delayed F1 is 0.4866; the seed-42 coin baseline has F1 0.3983. No-skill AP is prevalence, 0.3215. Insight: F1 is not accuracy, and 0.50 is not a universal random-F1 benchmark.

C6 | Classification report and confusion matrix. For selected LR/B:

Actual outcome	Predicted on time	Predicted delayed
On time: 6,785	TN = 1,958	FP = 4,827
Delayed: 3,215	FN = 244	TP = 2,971

Recall = TP/(TP+FN) = **92.41%**; precision = TP/(TP+FP) = **38.10%**; F1 = 2PR/(P+R) = **0.5395**. Accuracy is 49.29%. It catches most delays, but warns on 7,798/10,000 flights and **61.9% of warnings are false**.

C7 | Precision-recall curves and scoring times. Average precision (AP) summarizes precision across recall levels; LR/B has the highest AP here (0.5665). The curves describe ranking across thresholds, not a new warning policy. B prediction time for 10k flights: LR 0.023 s, KNN 11.69 s, SVM 13.88 s, RF 0.094 s. Scores are uncalibrated; SVM outputs margins, so do not describe every score as a true probability.

C8 | O2: paired A-to-B gains. F1 gains: **LR +0.0383, KNN +0.0322, SVM +0.0476, RF +0.0023**. LR's 95% day-bootstrap interval is [0.0330, 0.0443]; RF's [-0.0033, 0.0079] crosses zero. LR AP rises +0.0623. Richer information helps the first three clearly on F1 in this month; RF F1 improvement is uncertain. The 300-repeat bootstrap uses 31 dates and fixed fits. A/B measures the combined addition, not the separate effect of weather, history or traffic.

C9 | O3: subgroup error table. ORD precision is **32.2%** versus ATL 41.3% and DFW 41.6%. Morning recall is 68.5%; evening recall reaches 100% alongside **99.7% false-positive rate**, showing an overly broad warning policy. Some airline and overnight rates stay blank because denominators are too small. FPR = FP/(FP+TN); it differs from the false share among warnings, FP/(TP+FP).

C10 | O4: priority table and gains curve. Rank scores and review fixed 5%, 10% or 20% quotas; the dashed random line has expected recall equal to the budget.

Review quota	Reviewed	Delays found	Precision	Delay recall	Lift
5%	500	359	71.8%	11.2%	2.23x
10%	1,000	688	68.8%	21.4%	2.14x
20%	2,000	1,222	61.1%	38.0%	1.90x

At 10%, the historical-risk baseline finds 442 delays versus LR's 688. Lift = review precision / overall delay rate. Ranking helps concentrate risk within a limited capacity, even though the broad warning threshold produces many alerts.

C11 | Save and conclude. Saves metrics, scores, subgroup and priority outputs. The strongest practical finding is **risk prioritization**. This one summer month at three hubs cannot establish year-round/live reliability. Weather delivery, outcome access and schedule availability are assumptions; threshold transfer after refitting may affect alerts. The study does not measure money saved or delays prevented.

Sources: notebook 04, C1-C11; reports/final_models/RESULTS.md. Figures explained in prose.